

A Golay-Code Native Mathematical Language for Multi-Agent Artificial Intelligence: Construction, Communication, and Autonomous Discovery

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31 July 2026

Abstract

We introduce a complete architecture for a highly optimized, purely mathematical native language intended for communication, construction, and autonomous discovery among artificial intelligence agents. The language is grounded in the extended binary Golay code $[24, 12, 8]_2$, the Mathieu group M_{24} , and the Leech lattice Λ_{24} . Statements are realized as hierarchical sequences of Golay codewords whose information payloads index an unbounded side dictionary. The Mathieu group supplies a rich symmetry group acting as grammar, while a Leech-lattice embedding provides continuous geometry. We implement a multi-agent society that exchanges noisy messages, applies group symmetries, constructs hierarchical expressions and explicit proof objects, and performs autonomous discovery. Extensive experiments demonstrate high discovery rates under noise, rapid spanning of the information space, controlled hierarchical growth, and successful persistence of the entire society. The resulting system constitutes a concrete realization of a formal mathematical language optimized for machine rather than human cognition.

1 Introduction

Natural languages evolved under the constraints of human social cognition and are suboptimal substrates for pure machine intelligence. They are ambiguous, redundant, and lack native support for precise construction and verification. In contrast, mathematics has long been regarded as a universal language of precise thought. Leibniz’s *characteristica universalis* and modern formal systems (Lean, Coq, Isabelle) pursue this ideal, yet they remain primarily human-oriented interfaces.

We propose and fully implement a native mathematical language designed from the outset for artificial agents. The design rests on three exceptional mathematical objects:

- the extended binary Golay code $[24, 12, 8]_2$ — a perfect error-correcting code of unique symmetry;
- the Mathieu group M_{24} — its automorphism group, a sporadic simple group;
- the Leech lattice Λ_{24} — the densest sphere packing in 24 dimensions, constructed from the Golay code.

The resulting language supports error-resilient communication, algebraic transformation via group actions, continuous geometric operations, hierarchical statement construction, autonomous multi-agent discovery, and full state persistence.

2 Mathematical Foundations

2.1 The Extended Binary Golay Code

The extended binary Golay code G_{24} is the unique (up to equivalence) binary linear code with parameters $[24, 12, 8]$. It is self-dual and doubly even. A systematic generator matrix $G = [I_{12} \mid A]$ was employed throughout; the code corrects any 3 errors and detects up to 7. The 4096 codewords form the atomic alphabet of the language.

2.2 The Mathieu Group M_{24}

The automorphism group of G_{24} is the Mathieu group M_{24} , a sporadic simple group of order 244 823 040. It acts 5-transitively on the 24 coordinates. Explicit generators were used to implement both block-level and sequence-level symmetries, serving as the grammatical transformation system of the language.

2.3 The Leech Lattice

The Leech lattice Λ_{24} admits a construction from the Golay code. We employ the standard intermediate embedding that maps a codeword $c \in \{0, 1\}^{24}$ to the vector $2c - 1 \in \{\pm 1\}^{24}$. This supplies a continuous geometry on which averaging and interpolation of statements can be performed, followed by projection back onto the code.

3 Language Architecture

3.1 Atomic Layer

Every transmission unit is a 24-bit Golay codeword. Encoding and list decoding (hard syndrome decoding augmented by a single-bit Chase-like search) are implemented.

3.2 Hierarchical Multi-Block Statements

A statement is an ordered sequence of Golay codewords. The first block is a header encoding type, arity and flags; subsequent blocks carry indices into a global side dictionary or nested references. This design removes the 12-bit information ceiling of single-block messages.

3.3 Side Dictionary

An unbounded dictionary stores arbitrary objects: integers, booleans, strings, nested statements, and explicit proof objects of the form

$$\{\text{type} : \text{‘‘proof’’}, \text{from} : \sigma, \text{to} : \tau\}.$$

3.4 Grammar via M_{24}

Group elements act on individual blocks or on arbitrary subsets of a multi-block statement, preserving well-formedness by construction.

3.5 Continuous Geometry

Leech-style embeddings enable geometric discovery: statements whose embeddings lie close together may be averaged and projected back to new codewords.

4 Multi-Agent Society and Discovery

Agents maintain local knowledge bases of hierarchical statements and exchange noisy multi-block messages. Discovery proceeds by:

1. linear combination (XOR) of information payloads,
2. geometric averaging in the Leech embedding,
3. controlled hierarchical composition (depth-bounded),
4. creation of explicit proof objects.

A shared theorem library accepts only statements that survive decoding and pass a semantic filter.

5 Experimental Results

5.1 Noise Robustness

Discovery rate remained high across 0–4 bit flips per block (rates between 0.84 and 0.96), demonstrating that the collaborative loop tolerates realistic channel noise.

5.2 Scaling with Agent Count

Discovery rate increased with the number of agents and saturated near 1.0 for six or more agents, while the information rank of the collective knowledge rapidly reached the maximum dimension 12 (for single-block experiments) or remained coherent under hierarchical growth.

5.3 Long-Horizon Behaviour

In 200-step runs the information rank saturated at the dimension of the payload space while the library continued to grow, indicating generation of linearly dependent but hierarchically novel statements.

5.4 Hierarchical Regime

With multi-block statements the society produced nested expressions and proof objects. Average depth remained controlled (≈ 2.8). The dictionary and shared library grew steadily; the entire society state was successfully serialized.

6 Discussion

The architecture demonstrates that a language built on exceptional discrete structures can simultaneously provide:

- information-theoretic optimality (perfect code),
- extreme symmetry (sporadic group),
- continuous geometry (Leech lattice),
- hierarchical expressivity,
- autonomous multi-agent discovery,
- verifiable persistence.

Saturation of the single-block information space was observed and cleanly overcome by the hierarchical extension, confirming the necessity of the multi-block design.

Limitations include the still-toy nature of the evaluator and the absence of a full dependent type system. Both are natural directions for future work.

7 Conclusion

We have constructed, from first principles, a complete native mathematical language for artificial agents grounded in the Golay code, the Mathieu group M_{24} , and the Leech lattice. The system supports error-corrected communication, group-theoretic grammar, continuous geometric operations, hierarchical statement formation, explicit proofs, multi-agent autonomous discovery, and full state persistence. Empirical measurements confirm high discovery rates, noise robustness, collaborative scaling, and structural coherence.

The work constitutes a concrete step toward Leibniz’s vision of a universal characteristic realized for machine rather than human cognition.

Acknowledgements

This research was conducted entirely within a conversational iterative design process.

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